

ALPHA-HP-200 Hydraulic Press - Equipment Manual

Field	Value
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Tenant	Alpha Manufacturing (tenant-alpha)
Site	Ridgeway Works, Building 2 (metric documentation)
Document identifier	ALPHA-MAN-HP200-A
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Equipment	ALPHA-HP-200
Also covers	ALPHA-HP-200B (variant differences listed under Variant coverage)
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Applies to	Assets listed under equipment
Related documents	ALPHA-SAF-HP200-LOTO-1, ALPHA-PM-HP200-A, ALPHA-DIA-P100-1
Document owner	Maintenance Engineering

Purpose

This manual describes the design, rated limits and safe operation of the ALPHA-HP-200 four-post hydraulic press (serial range 2004-1xx) installed in Bay 2 of Ridgeway Works.

It is the controlled reference that maintenance technicians use when they build a work order for this press. Every set point, torque and interval stated here is authorised for this press only. Values taken from another press, from the as-built drawings, from a vendor brochure or from memory are not authorised values and must not be substituted.

This manual does not cover the following equipment:

ALPHA-HP-210, the 1000 kN bench press. It has a different frame, a different relief valve set point and a different guard arrangement. It has its own manual, AM-MAN-HP210.

ALPHA-HP-200B, an ALPHA-HP-200 with the 300 mm extended ram kit fitted. The differences are listed under Variant coverage. A work order that ignores those differences is a defective work order.

Equipment overview

The press is a four-post, single-ram unit fed from hydraulic power unit ALPHA-P-100 (30 kW, fixed displacement pump, 63 l/min). The ram is located and guided by four tie rods and four adjustable gib blocks; ram parallelism is maintained by adjusting the gib blocks, not by the hydraulic system.

Tooling is supported on an upper bolster carried by the ram and a fixed lower bolster. Both bolsters are T-slotted on 36 mm centres. Tool holding is by bolster clamping screws; the press has no die cushion, so the tool and the closed ram are left under full stored load whenever the ram is down. The counterbalance cylinders hold the ram weight only while the pump is running.

Control is a hard-wired pendant with a monitored two-hand control, a safety relay cabinet in panel 3A-04 and a mechanical die height sensor on the ram.

There is no closed-loop position control; ram position is derived from the sensor, not from the valve.

Technical data

Parameter	Value
Rated press force at full daylight	2000 kN
Relief valve set point (main circuit)	240 bar
Maximum permissible working pressure	235 bar
Hard-wired high pressure trip	250 bar
Secondary (tool clamp) circuit set point	180 bar
Rated capacity daylight	1200 mm
Stroke	800 mm
Die closing height	120 mm minimum, 1100 mm maximum
Bed size	1400 mm x 1200 mm, T-slots on 36 mm centres
Ram speed, approach	60 mm/s
Ram speed, working	25 mm/s
Ram speed, final 25 mm of stroke	10 mm/s maximum
Motor	30 kW, 400 V, three phase, 1460 rpm
Pump	fixed displacement, 63 l/min
Hydraulic oil	ISO VG 46 mineral, Alpha specification AM-H46
Reservoir	260 l, sight gauge plus high and low level switches
Return line filtration	10 micron, element AM-FL-10
Tie-rod nut torque	340 N.m, re-torque every 4000 running hours
Die height sensor calibration	every 1000 running hours
Hydraulic oil analysis	every 500 running hours
Ram parallelism	0.5 mm maximum per metre of die face
Mass	4200 kg
Noise at operator position	82 dB(A) during working stroke

Running hours are read from the hour meter at the pendant. The hour meter counts only while the pump motor is energised; it does not count standby time.

Operating conditions

Ambient air temperature: 5 degC to 35 degC. The press must not be started with oil temperature below 15 degC; warm the oil with the pump in bypass. Hydraulic oil temperature: 20 degC to 55 degC continuous. Above 60 degC the oil viscosity drops below the value the seals depend on.

The press is rated for indoor, non-corrosive service. It must not be used outdoors or in a wash-down area.

The press may be used for pressing, coining, rivet setting and plastic compression up to 120 degC tool temperature. Hot forging, welding and any operation that puts sparks near the ram seals is prohibited.

Maximum tool mass per station is 250 kg. Heavier tools must be handled with the overhead crane ALPHA-OC-5T and a spreader beam; the bolster clamp screws must not be used to lift anything.

Maximum combined tool height is limited by the 1100 mm die closing height.

A tool set-up that leaves less than 120 mm die closing height must be re-calculated by Maintenance Engineering before use.

Safeguards and control system

Two-hand control: monitored, safety-rated, mounted so that both hands must

be detected within the same 600 ms window. The minimum hold-to-run time is 600 ms; releasing either hand before 600 ms must not produce any ram motion. Safety-rated stop time, controls to ram stop: 250 ms maximum. Performance level of the two-hand control and the stop function: category 3, PL d, per the press safety circuit assessment. Upper ram stop switch and lower ram stop switch are independent of the die height sensor. Die height sensor: mechanical plunger type, part AM-DHS-22. It is a protective measure, not a positioning device. Counterbalance pressure must be present before the main relief opens. Loss of counterbalance with the ram down causes the ram to drop. The two-hand control must never be defeated, jumpered, taped over or bridged. Any diagnostic that requires defeating the safety circuit needs a documented risk assessment, a permit and a second authorised person present.

Operating limits and permitted range

1. Working pressure must not exceed 235 bar at the manifold. The relief valve is set at 240 bar and is not adjustable in service. Any requirement to reduce pressure is raised as a work order for Maintenance Engineering.
2. Pressing force is limited to 2000 kN at 1200 mm daylight. The force falls off as the die closes, because the pressure transducer used for the displayed force is calibrated for full daylight. Do not use the displayed force as a control value during a low daylight operation.
3. Point loading outside the bolster T-slots is not permitted. The load must be spread over the bolster area.
4. The ram must not be left resting on a tool with the pump stopped unless the tool is mechanically locked, because a loss of counterbalance with the pump stopped allows the ram to descend.
5. Ram parallelism must be within 0.5 mm per metre of die face before a precision operation. An operation that would damage the tool if the press is out of parallelism must not start until the ram parallelism check in Maintenance and inspection requirements is complete and recorded.

Routine operator checks before use

Performed by the operator, in this order, before the first cycle of a shift:

1. Check the reservoir oil level on the sight gauge. The oil must be between the two marks. A low level is a stop condition, not a top-up condition.
2. Check that there is no oil on the floor under the press, the power unit or the manifold. Any leakage is reported before the press is used.
3. Check that the tie rods, gib blocks, bolsters and the ram crown are free of visible damage and that no tools or scrap are inside the die area.
4. Check the two-hand control: operate it with the press in manual, confirm the safety relay reports a healthy chain in the diagnostics menu.
5. Confirm the die height sensor reading against the mechanical position indicator on the ram with the ram at the upper stop.
6. Run three complete cycles with the die area empty and observe the ram for side movement. More than 2 mm of side movement is a stop condition.
7. Check the audible warning and the emergency stop at the pendant.

Any failed check means the press is not released for production. Raise a work order and apply the isolation requirement in ALPHA-SAF-HP200-LOTO-1 if the ram, the bolsters or the hydraulic circuit have to be touched.

Variant coverage

ALPHA-HP-200B is an ALPHA-HP-200 retrofitted with the 300 mm extended ram kit (kit AM-KIT-R300). If the asset plate on the press carries the suffix B, the following values differ from Technical data and Variant coverage in this manual:

Parameter	ALPHA-HP-200	ALPHA-HP-200B
Rated daylight	1200 mm	1500 mm
Stroke	800 mm	800 mm
Tie-rod nut torque	340 N.m	390 N.m
Tie-rod re-torque interval	4000 running hours	3000 running hours
Bolsters required	standard HP-BOL-12	HP-BOL-15
Pressing force at full daylight	2000 kN	2000 kN

Using the 340 N.m figure on an HP-200B is a recognised cause of tie-rod failure. If the asset plate is missing or unreadable, stop and raise a work order; do not guess the variant.

ALPHA-HP-210 is not a variant of this press and nothing in this manual applies to it. See AM-MAN-HP210.

Maintenance and inspection requirements

The full preventive maintenance programme is in AM-PM-HP200. The items below are the limits from this manual that the programme must satisfy:

Hydraulic oil analysis every 500 running hours.
Tie-rod re-torque every 4000 running hours.
Die height sensor calibration every 1000 running hours.
Ram parallelism check every 2000 running hours.
Structural inspection of the four post welds every 6000 running hours.

Work on the hydraulic circuit, the bolsters, the tie rods or the ram requires the zero-energy state defined in ALPHA-SAF-HP200-LOTO-1 first. There is no task in this manual that may be started while the circuit is charged.

Documentation gaps

The following values are not contained in this manual. They are recorded on the as-built documentation held by Maintenance Engineering and that documentation is not part of this document collection:

Gib block and wedge clearance setting for this serial number: recorded on commissioning sheet CM-HP200-01.
Manifold fitting torque values: on the fitting vendor data sheets, not in this document.
Accelerometer alarm thresholds used by the optional ram monitoring system: set in the condition monitoring configuration, not in this manual.
Nitrogen pre-charge pressure of accumulator AC-1: as-built, see ALPHA-SAF-HP200-LOTO-1 required sequence, accumulator bleed-down step, for the bleed-down requirement.

If a technician needs one of these values and cannot read it from the controlled documents available at the machine, the correct action is a work order with documentation status "insufficient information", not a measured substitute.

Escalation conditions

Escalate immediately, before continuing the diagnosis, when any of the following occurs:

- Hydraulic pressure above 235 bar with no tooling load in the die.
- Audible knocking from the pump that does not stop when the relief opens.
- Visible hydraulic oil in the die area, on the tie rods or inside the frame.
- Ram side movement above 2 mm, or a parallelism reading outside 0.5 mm per metre.
- Any broken wire, cracked bolster clamp or damaged guard discovered during an inspection.
- Repeated tripping of the safety circuit on the same cycle of the two-hand control.
- A required value is missing from the controlled document set.

Escalation route: Level 2 Maintenance Engineering, then the Production Supervisor, then EHS for anything involving stored energy or structural damage.

Appendix: Recommended spares and consumables

The following spares are held for the ALPHA-HP-200 press. Quantities are the minimum stock for a single unit; increase them if a second press shares the same hydraulic power unit. Consumables are listed with the approved specification so that a substitute is never fitted without engineering approval.

Item	Part or specification	Minimum stock	Replacement interval
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Main relief valve cartridge	RV-200-A, 240 bar setting	1	On failure or at overhaul
Counterbalance valve	CB-200, pilot ratio 4.5:1	1	On failure
Ram seal kit	SK-200-80, nitrile and PTFE	2	At 8000 running hours or on leak
Gib wear strip set	GW-200, bronze faced	1 set	At 12000 running hours
Suction strainer element	63 micron, 63 l/min	2	Every 1000 running hours
Return filter element	10 micron absolute	3	Every 500 running hours
Accumulator bladder	AC-1, 6 l, nitrogen	1	Every 5 years
Pressure transducer	0 to 400 bar, 4 to 20 mA	1	On calibration failure
Two-hand control relay	Category 4 safety relay	1	On proof-test failure
Limit switch	IP67, roller lever	2	On failure
Hydraulic oil	ISO VG 46, AM-H46	520 l	Every 2000 hours or on analysis
Way lubricant	ISO VG 68	20 l	Top up as required

Spares fitted must be recorded on the work order with the part number and the asset serial number. A spare whose specification differs from the table is a non-conformance and must be raised with Maintenance Engineering before fitting.

Appendix: Commissioning and first-start checks

This appendix lists the checks required after any work that opens the hydraulic circuit, disturbs the frame or replaces a safety device. The checks are performed in order and the measured value is recorded, not a pass or fail mark. Work that has not been through this sequence is not returned to production.

1. Confirm the reservoir is filled to the mid to high mark with ISO VG 46 oil and that the suction strainer and return filter are new or proven.
2. Confirm every frame fastener is torqued to the values in the technical data table and that the tie-rod nuts are re-torqued if the frame was disturbed.
3. Confirm the accumulator AC-1 is charged to its specified nitrogen pre-charge and that the safety block is open.
4. Run the pump with the relief valve backed off and confirm the pump develops pressure smoothly with no cavitation noise. Record suction pressure.
5. Set the main relief valve to 240 bar using a calibrated gauge at test point TP-3. Record the lift-off and reseal pressure.
6. Confirm the hard-wired high-pressure trip operates at 250 bar and that the circuit cannot be reset below 230 bar.
7. Confirm the two-hand control holds for 900 ms and that releasing either button stops the ram within the documented stopping performance.
8. Confirm the light curtain and guard interlocks stop the ram when broken and that the press cannot be restarted until the guard is closed and reset.
9. Cycle the press ten times at working speed and confirm the ram stops at the bottom of the stroke without drift for five minutes.
10. Record oil temperature at the end of the run and confirm it is below the alarm set point.

Every value is entered on the commissioning record and countersigned by the person who authorized the work.

Revision history

Revision	Effective date	Summary
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Rev A	2023-03-01	First controlled issue for ALPHA-HP-200 after the 2023 press rebuild.